FOOD DELIVERY OPERATIONS

Complete Data Analytics Project | Kagnus

Final evidence-based analysis report

Executive summary

This project reviews 45,584 food-delivery records with 20 source columns. The mean recorded delivery duration is 26.29 minutes (median 26.00). The analysis documents data-quality limitations, compares operational categories, and provides six charts. No causal claims are made.

Data source and approach

Source: Kagnus-provided Food Delivery Dataset. Analysis used Python, Pandas, NumPy, Matplotlib and Seaborn. The original dataset is preserved separately; the cleaned file and cleaning log are supplied in the repository. One row represents a delivery record.

Data quality and verification

Original rows: 45,584. Cleaned rows: 45,584. Exact duplicate rows: 0. Duplicate delivery IDs: 0. Rows were not automatically removed for missing values or potential outliers. See docs/cleaning_log.csv for decisions.

Column	Missing	% of rows
Delivery_person_Ratings	1908	4.19%
Delivery_person_Age	1854	4.07%
Time_Orderd	1731	3.80%
City	1200	2.63%
multiple_deliveries	993	2.18%
Weather_conditions	616	1.35%
Road_traffic_density	601	1.32%
Festival	228	0.50%

Evidence-backed observations

Traffic: Jam mean 31.18 min (n=14,139); Low mean 21.27 min (n=15,476). Difference 9.91 min.

Weather: Cloudy mean 28.92 min (n=7,533); Sunny mean 21.86 min (n=7,282). Difference 7.06 min.

Vehicle: motorcycle mean 27.61 min (n=26,429); electric_scooter mean 24.47 min (n=3,814). Difference 3.14 min.

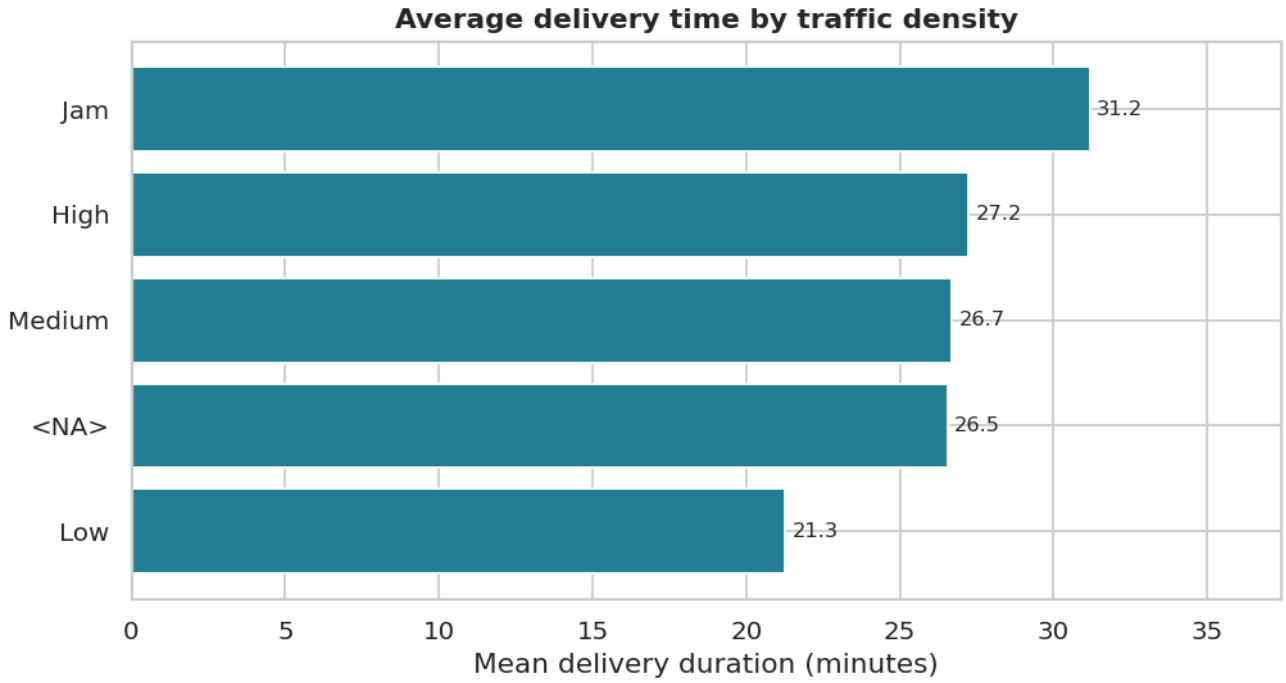
City: Semi-Urban mean 49.73 min (n=164); Urban mean 22.98 min (n=10,133). Difference 26.75 min.

Multiple: 3.0 mean 47.82 min (n=361); 0.0 mean 22.88 min (n=14,094). Difference 24.94 min.

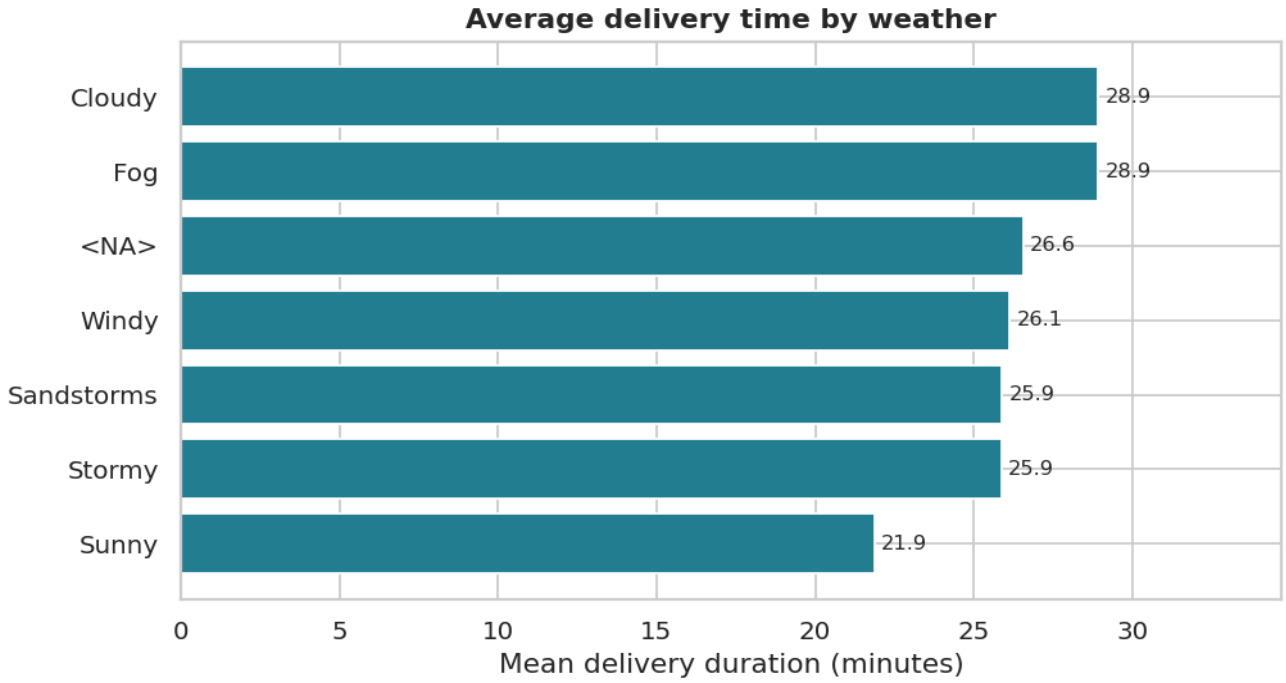
Order: Meal mean 26.42 min (n=11,456); Drinks mean 26.19 min (n=11,321). Difference 0.23 min.

Operational visualizations

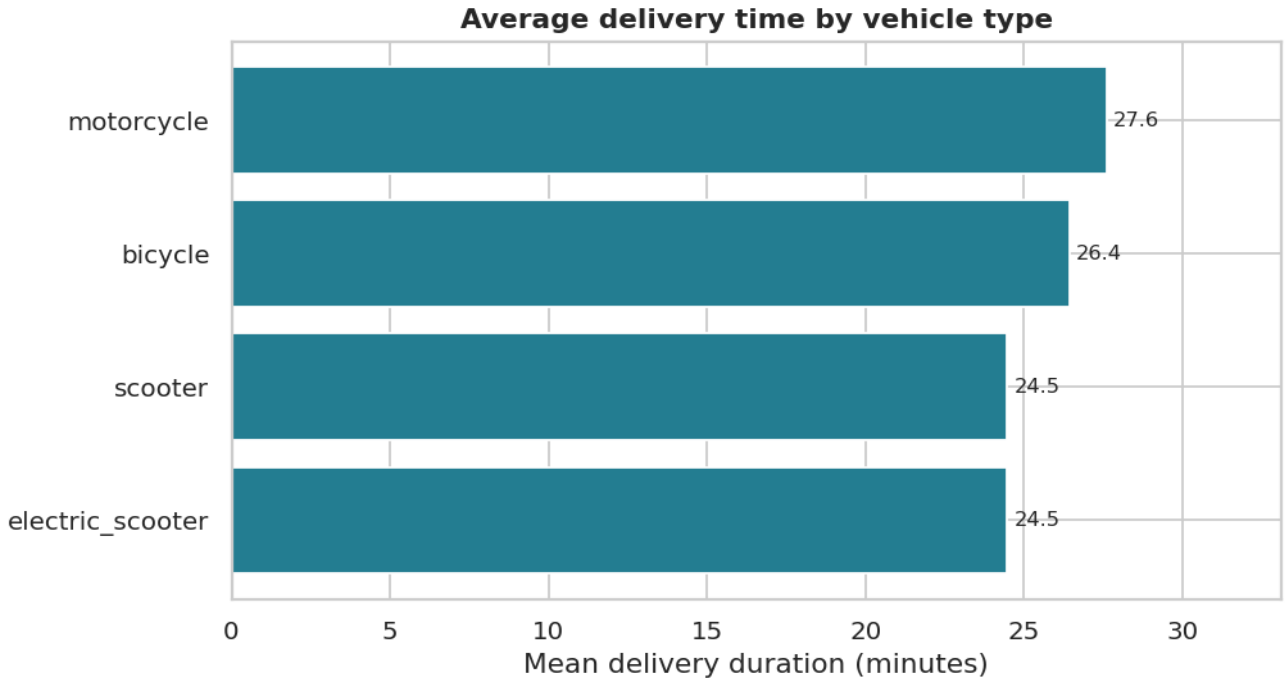
Average delivery time by traffic density



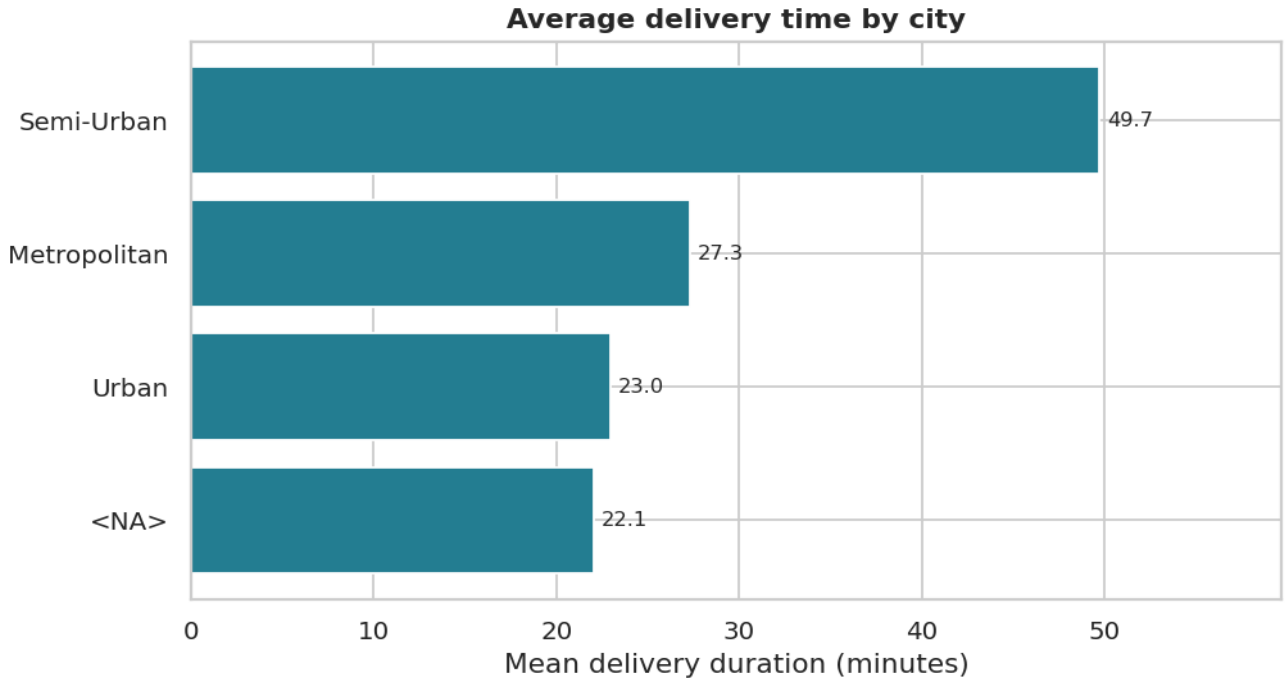
Average delivery time by weather



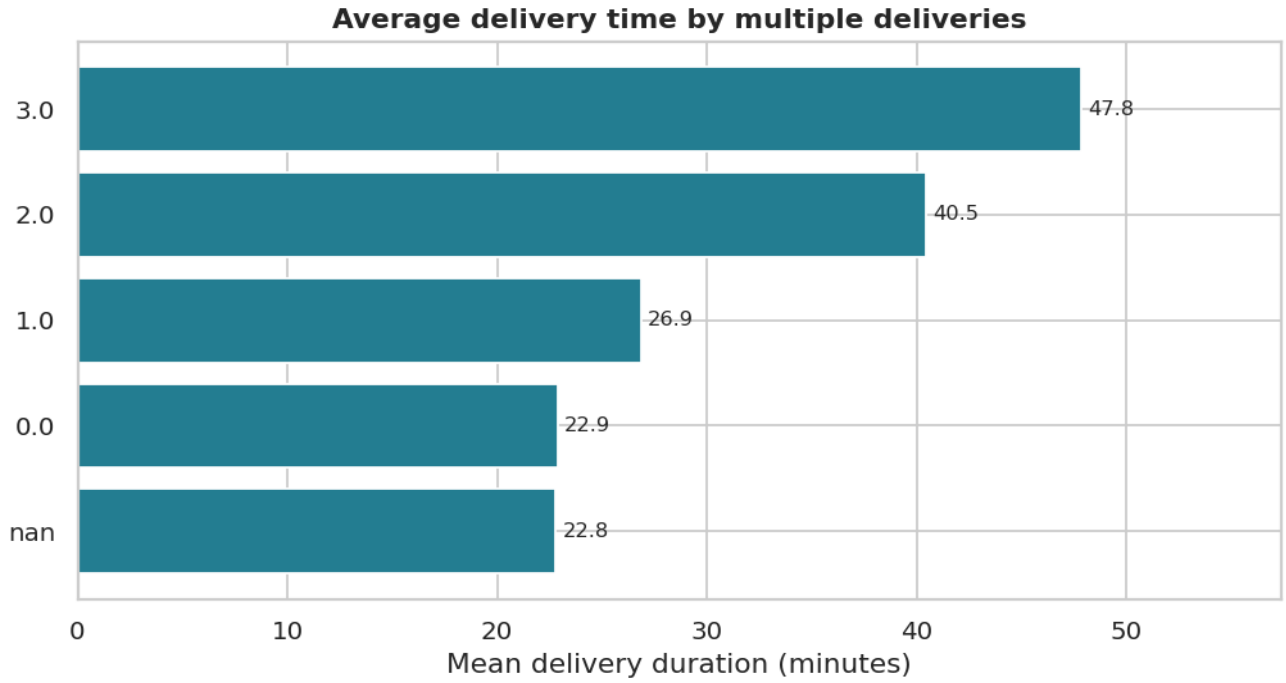
Average delivery time by vehicle type



Average delivery time by city

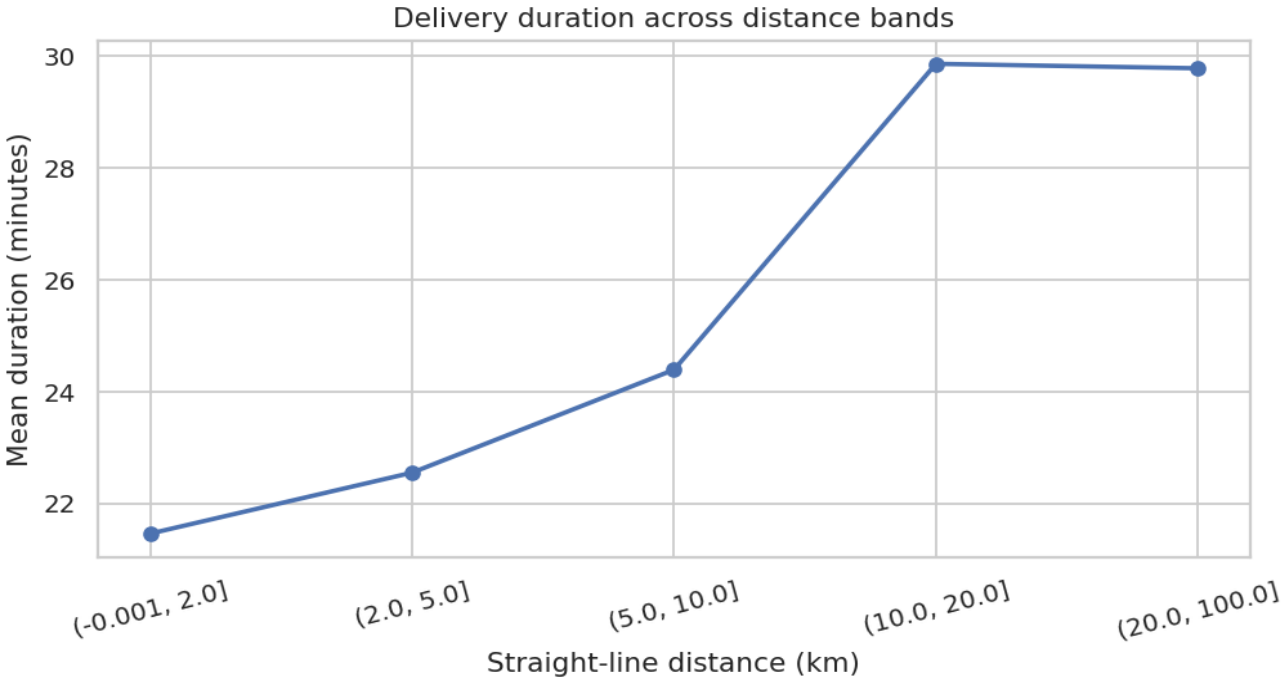


Average delivery time by multiple deliveries



Distance-based cross-check

Straight-line distance was computed for 41,513 records with plausible coordinates and distance at most 100 km. This is not road-route distance. The exclusion rule applies only to distance comparisons.



Recommendations

• Monitor traffic and weather categories with longer observed durations to prioritize operational review.

• Evaluate vehicle and city patterns after accounting for distance and traffic; do not assume category differences are causal.

• Improve capture of age, ratings, city, and order times where missing.

• Validate anomalous ratings and coordinates with the data owner before correcting them.

• Pilot routing and scheduling changes and measure impact before scaling.

Limitations and conclusion

Missing values, undocumented codes, questionable coordinates, and varying category sizes constrain interpretation. The duration definition is not fully documented, and the data are historical. The analysis establishes a reproducible baseline for operational monitoring, not a causal model. Detailed executable calculations, cleaning decisions, and summary CSVs are available in the GitHub project.